# A/B TEST

## Analysis

Findings & Write-up

*Lawrence | Data Analyst Portfolio*

# About the Data

## Dataset

Kaggle — Marketing A/B Dataset (andrewjayasatyo/ab-testing-ad-vs-psa-effectiveness)

## Overview

NutriVance, a health and wellness e-commerce brand, is evaluating whether paid advertisements or public service announcements are more effective at driving customer conversions. With limited marketing budget, the business needs to determine whether ad spend is generating a statistically significant lift in conversions before committing to either channel as its primary marketing approach. This analysis applies a chi-square test to evaluate whether the difference in conversion rates between the two groups is statistically significant, and examines ad exposure volume, day, and hour patterns to surface additional targeting insights.

# Data & Experiment Setup

## Key Columns

| Column Name | Description |
|---|---|
| user id | Unique identifier for each user |
| test group | Group assignment — 'ad' (advertisement) or 'psa' (public service announcement) |
| converted | Whether the user converted — True or False |
| total ads | Total number of ads shown to the user |
| most ads day | Day of the week the user was shown the most ads |
| most ads hour | Hour of the day the user was shown the most ads |

## Custom Columns

| Column Name | How It Was Calculated |
|---|---|
| exposure volume | Users bucketed into Low, Mid, and High Exposure tiers based on total ads using pd.cut() |

## Group Definitions & Sizes

| Group | Definition | Size |
|---|---|---|
| ad | users who were shown a paid advertisement | 564577 |
| psa | users who were shown a public service announcement | 23524 |

### Metric Measured

**Conversion rate — the proportion of users in each group who converted (converted = True).**

### Note on Group Size Imbalance

**The group size between ad and psa groups has a ratio of 24:1 (564577 & 23524) which is overly imbalanced but does not affect the chi-square results as all expected cell values remain well above the minimum threshold of 5, with the smallest at approximately 594. The smaller group is also large enough that the variance would be negligible.**

## Statistical Test Results

### Conversion Rates per Group

| Group | Conversion | Total Users | Conversion Rate |
|---|---|---|---|
| ad | 14423 | 564577 | 2.55% |
| psa | 420 | 23524 | 1.79% |

### Chi-Square Test

| Chi-Square Statistic | P-Value | Result |
|---|---|---|
| 54.005823883685245 | 1.9989623063390075e-13 | Reject null hypothesis. There is a significant difference |

The p-value of 1.999e-13 falls below the 0.05 threshold, which means there is a significant difference between the conversion rates and not due to chance.

### Effect Size — Cramér's V

| Cramér's V | Strength of Association |
|---|---|
| 0.0096 | Weak Association |

While the difference is statistically significant, a Cramér's V of 0.0096 indicates a weak association — meaning group membership (ad vs psa) has very little practical influence on whether a user converts. The effect is real but small.

### Confidence Interval (95%)

| Lower Bound | Upper Bound |
|---|---|
| 0.60% | 0.94% |

The confidence interval tells us that we are 95% confident the true difference in conversion rates between the ad and psa groups lies somewhere between 0.60% and 0.94% .

## Supporting Analysis

## Ad Exposure Volume vs Conversion Likelihood

| Exposure Level | Size | Conversion Rate |
|---|---|---|
| Low Exposure | 587909 | 2.52% |
| Mid Exposure | 184 | 16.85% |
| High Exposure | 8 | 50.00% |

The High Exposure bucket only has 8 users, making the 50% conversion rate statistically unreliable. Low and Mid Exposure tiers are the meaningful comparison.

## Conversion Rate by Day and Group

| Day | Group | Conversion Rate |
|---|---|---|
| Monday | ad | 3.32% |
| Monday | psa | 2.26% |
| Tuesday | ad | 3.04% |
| Tuesday | psa | 1.44% |
| Wednesday | ad | 2.54% |
| Wednesday | psa | 1.58% |
| Thursday | ad | 2.16% |
| Thursday | psa | 2.02% |
| Friday | ad | 2.25% |
| Friday | psa | 1.63% |
| Saturday | ad | 2.13% |
| Saturday | psa | 1.40% |
| Sunday | ad | 2.46% |
| Sunday | psa | 2.06% |

## Conversion Rate by Hour of Day

| Hour | Conversion Rate |
|---|---|
| 0:00 | 1.84% |
| 1:00 | 1.29% |
| 2:00 | 0.73% |
| 3:00 | 1.05% |
| 4:00 | 1.52% |

| | |
|---|---|
| 5:00 | 2.09% |
| 6:00 | 2.22% |
| 7:00 | 1.81% |
| 8:00 | 1.95% |
| 9:00 | 1.92% |
| 10:00 | 2.15% |
| 11:00 | 2.21% |
| 12:00 | 2.38% |
| 13:00 | 2.47% |
| 14:00 | 2.81% |
| 15:00 | 2.97% |
| 16:00 | 3.08% |
| 17:00 | 2.82% |
| 18:00 | 2.74% |
| 19:00 | 2.67% |
| 20:00 | 2.98% |
| 21:00 | 2.89% |
| 22:00 | 2.61% |
| 23:00 | 2.27% |

## Key Findings

- The ad group produced a statistically significant higher conversion rate than the PSA group at 2.55% vs 1.79%, with a p-value of 1.999e-13 confirming the difference is not due to chance.
- Cramér's V of 0.0096 indicates a weak association — which means that group membership has very little real-world influence on whether a user converts.
- The true difference between the ad and psa group conversion rate will fall between 0.60% and 0.94% with a confidence of 95%.
- The low exposure bucket has the highest size of 587k with a conversion rate of only 2.52% and then as the exposure increases, the conversion also increases with 16.85% and then 50% for mid and high exposure respectively. However, it’s too soon to say that more ads means higher conversion rate because the size of both mid and high exposure buckets (184 & 8) is too small to be a reliable indicator.

- The conversion rate for both groups starts the highest on Monday and then comes back again on Sunday. The only difference is that the psa group peaks on Thursdays unlike the ad group. The pattern suggests that Monday is the strongest day to run ads regardless of channel.
- The best performing windows for ads are 16 (late afternoon), and 20 (evening rise). With 16:00 having the highest peak at 3.08%, followed by 20:00 at 2.98%.

# Overall Recommendation

## Target Segment

*The mid exposure ad group, every monday and sunday, at 16:00 in the late afternoon and 20:00 in the start of the evening*

## Reasoning

This segment has shown to be the most promising when it comes to higher conversion rate. As shown by the statistical results there is a significant difference between the conversion rate of ads and psa. With ads producing a conversion rate of 2.55% compared to 1.79% for PSAs. The best timing of day and hour is also proven in the supporting analysis. The mid exposure group is the most promising target as its conversion rate of 16.85% significantly outperforms low exposure at 2.52%, though its smaller sample size means a larger dataset is needed to confirm this before fully committing to it.

## Suggested Actions

- Prioritize ad budget allocation over PSA spend
- Use PSAs as a low-cost supplementary channel while prioritizing ad spend
- Investigate optimal ad frequency once larger exposure samples are available.
- Develop an automatic posting of ads on Monday and Sunday, in the late afternoon and start of the evening